

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5	Assessment:	Assessment Tool 1
Weightage:	40% of CIA	Total Marks:	100
CO1 Statement:	CO5-AT1: Pseudocode Design		
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Section / Batch:	CSE(DS)	Date:	26/08/26

Code of Conduct

I **T. Niranjana Kumar (192473007)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: T. Niranjana

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. Supervised Learning

Question: Design pseudocode to build a supervised learning workflow that accepts a labelled dataset, separates features and target values, trains a model, and predicts the target for new data.

Pseudocode

```
BEGIN
INPUT labelled_dataset
SEPARATE features,target
TRAIN model
INPUT new_data
PREDICT target
PRINT prediction
END
```

2. kNN Classifier

Question: Design pseudocode for a k-Nearest Neighbors (kNN) classifier that calculates the distance between a test sample and all training samples, selects the k nearest samples, and predicts the class using majority voting.

Pseudocode

```
BEGIN
INPUT training_data,test_sample,k
CALCULATE distance to all samples
SELECT k nearest
PREDICT majority class
END
```

3. kNN Classifier

Question: Design pseudocode to classify a new student as Pass or Fail using kNN based on attendance percentage, internal marks, and assignment scores.

Pseudocode

```
BEGIN
INPUT attendance,marks,assignment
CALCULATE distances
SELECT k neighbors
IF majority=Pass THEN PRINT Pass ELSE PRINT Fail
END
```

4. kNN Classifier

Question: Design pseudocode for selecting the appropriate value of k in a kNN classifier by evaluating different k values on validation data.

Pseudocode

```
BEGIN
FOR k=1 TO MaxK
  TRAIN & VALIDATE
  STORE accuracy
ENDFOR
SELECT k with highest accuracy
END
```

5. Decision Tree

Question: Design pseudocode to construct a Decision Tree classifier by selecting the best feature for splitting the training dataset and recursively creating child nodes.

Pseudocode

```
BEGIN
IF all records same class RETURN leaf
SELECT best feature
SPLIT dataset
RECURSIVELY create child nodes
END
```

6. CART

Question: Design pseudocode for the CART classification algorithm to select the best binary split using Gini impurity and construct a classification tree.

Pseudocode

```
BEGIN
CALCULATE Gini for all splits
SELECT lowest Gini split
CREATE left/right child
REPEAT until stopping
END
```

7. Decision Tree

Question: Design pseudocode to predict whether a customer will purchase a product using a Decision Tree based on age, income, previous purchases, and browsing history.

Pseudocode

```
BEGIN
INPUT age,income,purchases,browsing
APPLY decision rules
PRINT Buy or Not Buy
END
```

8. Regression Analysis

Question: Design pseudocode for a regression analysis workflow that loads a dataset, identifies independent and dependent variables, trains a regression model, and evaluates its predictions.

Pseudocode

```
BEGIN
LOAD dataset
IDENTIFY X,Y
TRAIN regression model
PREDICT values
EVALUATE model
END
```

9. Linear Regression

Question: Design pseudocode to implement simple linear regression for predicting house price from house area using the least-squares approach.

Pseudocode

```
BEGIN
INPUT area,price
CALCULATE slope & intercept
PREDICT house price
END
```

10. Linear Regression

Question: Design pseudocode to implement multiple linear regression for predicting a student's final mark using attendance, internal marks, assignment marks, and study hours.

Pseudocode

```
BEGIN
INPUT attendance,internal,assignment,hours
TRAIN MLR model
PREDICT final mark
END
```

11. Linear Regression

Question: Design pseudocode to train a linear regression model, calculate predicted values and residuals, and determine the Mean Squared Error (MSE) and R^2 score.

Pseudocode

```
BEGIN
PREDICT values
CALCULATE residuals
COMPUTE MSE
COMPUTE  $R^2$ 
PRINT results
END
```

12. Logistic Regression

Question: Design pseudocode for logistic regression to predict whether a student will pass or fail based on study hours and attendance percentage.

Pseudocode

```
BEGIN
INPUT study_hours,attendance
TRAIN logistic model
PREDICT Pass/Fail
END
```

13. Logistic Regression

Question: Design pseudocode that calculates the sigmoid probability for logistic regression and converts the probability into a binary class using a threshold of 0.5.

Pseudocode

```
BEGIN  
CALCULATE z  
prob=1/(1+e-z)  
IF prob>=0.5 THEN Class=1 ELSE Class=0  
END
```

14. Logistic Regression

Question: Design pseudocode to evaluate a logistic regression classifier using the confusion matrix, accuracy, precision, recall, and F1-score.

Pseudocode

```
BEGIN  
CALCULATE TP,TN,FP,FN  
COMPUTE Accuracy,Precision,Recall,F1  
PRINT metrics  
END
```

15. Unsupervised Learning

Question: Design pseudocode for an unsupervised learning process that accepts an unlabelled dataset, preprocesses the features, discovers groups, and produces cluster assignments.

Pseudocode

```
BEGIN  
LOAD unlabelled data  
PREPROCESS features  
DISCOVER clusters  
OUTPUT assignments  
END
```

16. K-Means Clustering

Question: Design pseudocode to implement the K-Means clustering algorithm: initialize centroids, assign observations to the nearest centroid, update centroids, and repeat until convergence.

Pseudocode

```
BEGIN
INITIALIZE centroids
ASSIGN nearest centroid
UPDATE centroids
REPEAT until convergence
END
```

17. K-Means Clustering

Question: Design pseudocode to group customers into three clusters based on annual income and spending score using K-Means clustering.

Pseudocode

```
BEGIN
INPUT income, spending
SET K=3
RUN K-Means
PRINT cluster IDs
END
```

18. K-Means Clustering

Question: Design pseudocode to determine the suitable number of clusters using the Elbow Method and then apply K-Means with the selected number of clusters.

Pseudocode

```
BEGIN
FOR K=1 TO MaxK
  RUN K-Means
  STORE WCSS
ENDFOR
SELECT elbow K
END
```

19. Evaluation Metrics

Question: Design pseudocode to calculate TP, TN, FP, and FN from actual and predicted binary class labels and subsequently calculate accuracy, precision, recall, and F1-score.

Pseudocode

```
BEGIN
INPUT actual,predicted
CALCULATE TP,TN,FP,FN
CALCULATE Accuracy,Precision,Recall,F1
END
```

20. Model Evaluation

Question: Design pseudocode for comparing kNN, Decision Tree, Logistic Regression, and K-Means models using appropriate evaluation measures and selecting the most suitable model for a given Data Science problem.

Pseudocode

```
BEGIN
EVALUATE kNN,DT,LR,KMeans
COMPARE metrics
SELECT best model
PRINT result
END
```